Data, corrections and assessment

Objective and inputs

Predict **repeat_purchase_90d**: whether the customer purchases again within 90 days of the current purchase occasion. Each row is an occasion, not a customer. Train has 6,534 labelled rows and 39 columns (55.9% positive); test has 986 later unlabelled rows. The optional client table has 1,143 customers and 13 fields.

Pipeline and fit boundaries

Stage / notebook	Technique and rationale	Data boundary
1. Audit / 01 + 02	Describe purchases and clients; inspect missingness, target relationships and cross-column identities.	Raw files preserved.
2. Prepare / 03	Repair exact same-row identities; otherwise invalidate impossible history. Exclude 65 undated train rows.	No fitted statistics or target used.
3. Feature sets / 02 + 03	Transactions only versus + country, commercial region, island and categorical entry year; validated customer join.	Same purchase IDs in both branches.
4. Fit / 03	Training medians + residual missing indicators; StandardScaler; Missing category; one-hot; optional L1; logistic classifier.	Fit on each training partition only.
5. Select and assess / 03	Inner F1 chooses L1/noise settings; main validation compares the two feature sets.	Chronological splits; both have 90-day gaps.
6. Refit and export / 03	Refit the complete chosen rule on all eligible labelled rows; predict every test ID.	Test transformed only; no test fitting.

Audited corrections

Issue / exact recovery	Train	Test
Inconsistent historical mean	78	11
Missing mean recovered with positive count	194	30
Impossible historical maximum	48	7
Impossible annual spend	70	11
Distinct purchases with invalid cells	227	32

Rebuild mean as total/count only when count is positive. Recover four train maxima exactly (one prior purchase: total; two: total minus minimum); leave other invalid maxima missing. Equal-count nested windows recover 108/5 annual amounts, including missing values. Unrecoverable values and negative time intervals stay missing for fold-fitted imputation. Counts/totals/minima/spread and shorter windows are trusted after consistency checks. No rounding-by-default, future events or target-derived repairs are used.

Temporal design

Main training: **4,424** purchases, 2015-07-11 to 2022-04-10. A 90-day window starting 2022-04-14 excludes **103** purchases. Validation: **1,942**, 2022-07-13 to 2025-02-15. Complete training target windows finish before validation starts. Shared customers are allowed because the task includes their future purchases. Dates are anonymised but order and intervals are preserved.

Selection, evidence and final solution

Selection and modelling choices

Inside the 4,424 main-training rows, an inner temporal split uses 2,971 fitting rows, a 90-day gap of 111, and 1,342 validation rows starting 2020-07-04. Compare eight settings per feature set: no selection or L1 C in {0.01, 0.1, 1}, each with/without random_noise. Preprocessing and selection are refitted inside each candidate.

Embedded L1 is suitable for the chosen logistic family and related historical predictors. SelectFromModel retains coefficients with magnitude at least 1e-8. Choose the highest inner F1; exact ties prefer fewer encoded columns, then no noise. Smaller C means stronger selection. The final classifier stays L2 logistic regression (C=1, lbfgs, max_iter=10,000), with threshold 0.5. Standardisation prevents numeric units dominating regularisation; medians provide a simple outlier-resistant fill for residual numeric gaps.

Main validation results

Feature set	Selection	F1	Precision	Recall	ROC-AUC
Always positive	None	0.7211	0.5639	1.0000	0.5000
Transactions	None	0.7263	0.7963	0.6676	0.7794
Transactions + clients	None	0.7082	0.7952	0.6384	0.7742
Transactions	L1 C=0.01	0.7350	0.7613	0.7105	0.7816
Transactions + clients	L1 C=0.01	0.7338	0.7773	0.6950	0.7812

F1 is the supplied competition metric; precision, recall and ROC-AUC give context. Selected feature set: **Transactions**, F1 **0.7350**. The margin over the always-positive reference is 1.39 percentage points. Adding clients changes selected-procedure F1 by -0.12 points on this holdout. This small difference does not establish a general advantage or disadvantage of client data.

Noise, final features and predictions

The chosen rule is L1 C=0.01, with raw noise included as a candidate; random_noise is discarded by the final selection. This empirical decision is not a leakage test. Refit the complete pipeline on all 6,469 eligible labelled purchases, including the old gap and validation. The final selector retains 12 of 149 encoded columns, representing 12 original variables.

Final original variables: Total Imp_, Total Desc_ Linha, prior_purchase_count, prior_max_purchase_value, prior_mean_gap_days, purchase_count_90d, purchase_count_365d, spend_365d, prior_std_purchase_value, prior_std_gap_days, salesperson, commercial_zone.

Refitting can change selected columns. The export contains 986 binary predictions (585 positive), with original IDs, order and read-back checks. Test runs from 2025-05-24 to 2026-10-06; all training label windows finish before it starts. The predicted positive fraction is not test accuracy.

Limits and reproducibility

EDA used all labelled data and main scores informed the final feature-set choice: these are development results, not an independent final estimate. One inner/outer holdout cannot establish stability or statistical significance. Current client attributes are assumed historically available. The amount identities and nonnegative history support the correction rules but remain measurement assumptions. The fixed threshold was not optimised. All three notebooks, shared source, original CSVs, environment requirements and generated evidence accompany the submission.

Class basis: Feature Selection slides (embedded L1), Week 04 classification (SelectFromModel), and Week 05 model selection (fit within partitions; refit the chosen pipeline).